

MH248REST Hall-effect sensor is a temperature stable, stress-resistant, micro-power switch. Superior high-temperature performance is made possible through a dynamic offset cancellation that utilizes chopper-stabilization. This method reduces the offset voltage normally caused by device over molding, temperature dependencies, and thermal stress.

MH248REST includes the following on a single silicon chip: voltage regulator, Hall voltage generator, small-signal amplifier, chopper stabilization, Schmitt trigger, open-drain output. Advanced CMOS wafer fabrication processing is used to take advantage of low-voltage requirements, component matching, very low input-offset errors, and small component geometries.

This device requires the presence of omni-polar magnetic fields for operation.

MH248REST is rated for operation between the ambient temperatures -40°C and $+85^{\circ}\text{C}$ for the E temperature range. Package types ST is an SOT-23(0.8 mm nominal height)

The package type is in a lead Halogen Free version has been verified by third party Lab.

Features and Benefits

- CMOS Hall IC Technology.
- Solid-State Reliability.
- Micro power consumption for battery-powered applications.
- Omni polar, output switches with absolute value of North or South pole from magnet.
- Operation down to 2.5 V and Max at 5.0V.
- Ultra High Sensitivity for direct reed switch replacement applications.
- Narrower operating range(Bop distribution).
- Custom sensitivity selection is available in optional package.
- Pb Free/Green chip is qualified by third party lab.

Applications

- Solid state switch
- Handheld Wireless Handset Awake Switch (Flip Cell/PHS Phone/Note Book/Flip Video Set)
- Lid close sensor for battery powered devices
- Magnet proximity sensor for reed switch replacement in low duty cycle applications

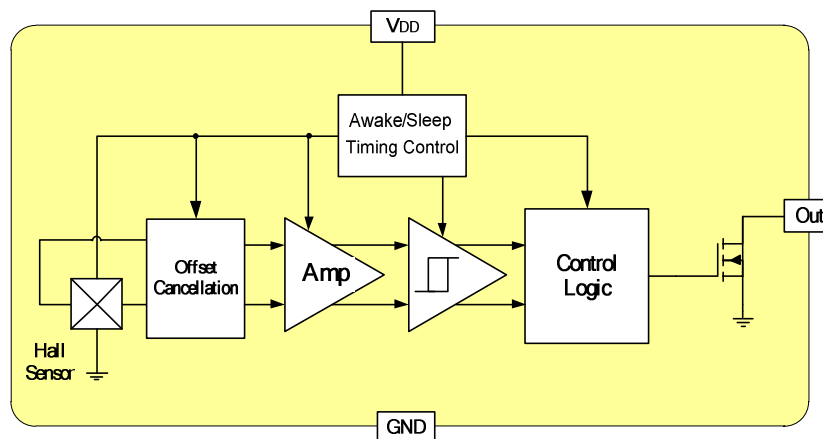
Ordering Information

	Company Name and Product Category
	MH:MST Hall Effect/MP:MST Power IC
	Part number
	181,182,183,184,185,248,249,276,477,381,381F,381R,382.....
	If part # is just 3 digits, the forth digit will be omitted.
	Temperature range
	E: 85 °C, I: 105 °C, K: 125 °C, L: 150 °C
	Package type
	UA:TO-92S,VK:TO-92S(4pin),VF:TO-92S(5pin),SO:SOT-23, SQ:QFN-3,ST:TSOT-23,SN:SOT-553,SF:SOT-89(5pin), SS:TSOT-26,SD:DFN-6
	Sorting
	α,β,Blank.....

Part No.	Temperature Suffix	Package Type
MH248REST	E (-40°C to +85°C)	ST(TSOT-23)

Custom sensitivity selection is available by MST sorting technology

Functional Diagram



Note: Static sensitive device; please observe ESD precautions. Reverse V_{DD} protection is not included. For reverse voltage protection, a 100Ω resistor in series with V_{DD} is recommended.

Absolute Maximum Ratings At($T_a=25^{\circ}\text{C}$)

Characteristics		Values	Unit
Supply voltage, (V_{DD})		6	V
Output Voltage, (V_{out})		6	V
Reverse voltage, (V_{DD}) (V_{OUT})		-0.3	V
Magnetic flux density		Unlimited	Gauss
Output current (I_{out})		2	mA
Operating temperature range, (T_a)		-40 to +85	$^{\circ}\text{C}$
Storage temperature range, (T_s)		-55 to +150	$^{\circ}\text{C}$
Maximum Junction Temp, (T_j)		150	$^{\circ}\text{C}$
Thermal Resistance	(θ_{JA}) ST	310 $^{\circ}\text{C}/\text{W}$	$^{\circ}\text{C}/\text{W}$
	(θ_{JC}) ST	223 $^{\circ}\text{C}/\text{W}$	$^{\circ}\text{C}/\text{W}$
Package Power Dissipation, (P_D) ST		400	mW

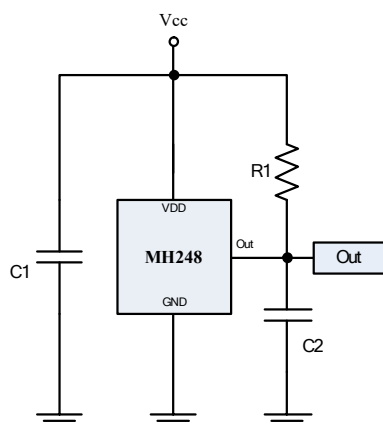
Note: Exceeding the absolute maximum ratings may cause permanent damage. Exposure to absolute maximum-rated conditions for extended periods may affect device reliability.

Electrical Specifications

DC Operating Parameters $T_A=+25^{\circ}\text{C}$, $V_{DD}=3.0\text{V}$

Parameters	Test Conditions	Min	Typ	Max	Units
Supply Voltage, (V_{DD})	Operating	2.5		5.0	V
Supply Current, (I_{DD})	Awake State		2.5	4.0	mA
	Sleep State		8.0	12	μA
	Average		10	16	μA
Output Leakage Current, (I_{off})	Output off			1	μA
Output Low Voltage, (V_{sat})	$I_{OUT}=1\text{mA}$			0.3	V
Awake mode time, (T_{aw})	Operating		70		μs
Sleep mode time, (T_{SL})	Operating		70		mS
Duty Cycle, (D, C)			0.1		%

Typical Application circuit



C1 : 10nF
C2 : 100pF
R1 : 100K Ω

MH248REST Magnetic Specifications

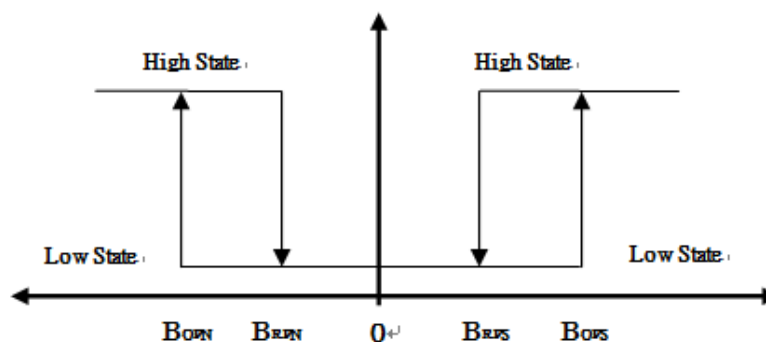
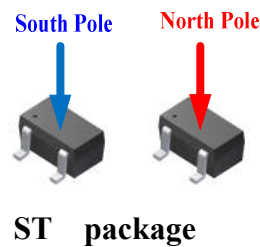
DC Operating Parameters $T_A = +25\text{ }^{\circ}\text{C}$, $V_{DD} = 3.0\text{V}$

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Units
Operating Point	B_{OPS}	N pole to branded side, $B > BOP$, V_{out} off	30		60	Gauss
	B_{OPN}	S pole to branded side, $B > BOP$, V_{out} off	-60		-30	Gauss
Release Point	B_{RPS}	N pole to branded side, $B < BRP$, V_{out} ON	20		50	Gauss
	B_{RPN}	S pole to branded side, $B < BRP$, V_{out} ON	-50		-20	Gauss
Hysteresis	B_{HYS}	$ BOPx - BRPx $		7		Gauss

MH248REST Output Behavior versus Magnetic Polar

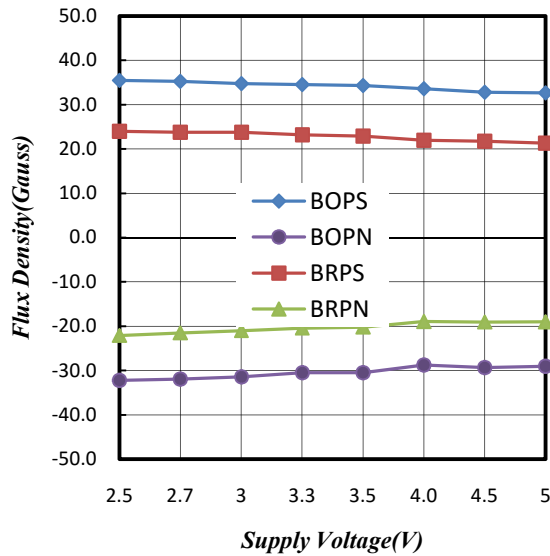
DC Operating Parameters $T_a = -40\text{ to }85\text{ }^{\circ}\text{C}$, $V_{DD} = 2.5\text{V to }5.0\text{V}$

Parameter	Test condition	OUT(ST)
South pole	$B < Bop[(-60 \sim (-30))]$	High
Null or weak magnetic field	$B = 0$ or $B < BRP$	Open(Pull-up Voltage)
North pole	$B > Bop(30 \sim 60)$	High

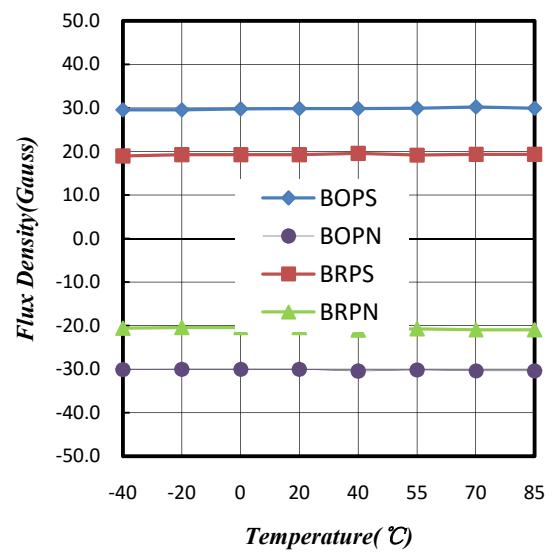


Performance Graph

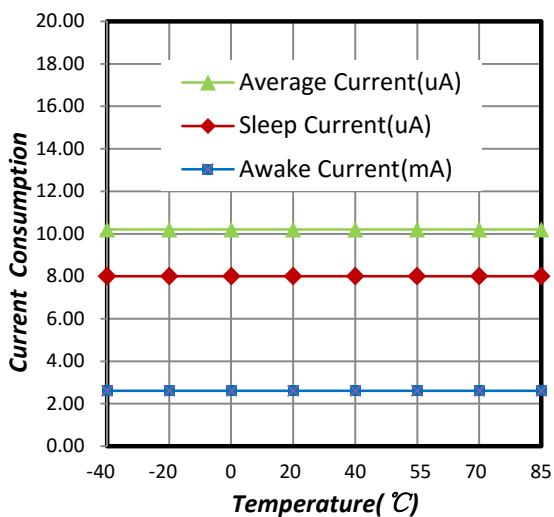
Typical Supply Voltage(V_{DD}) Versus Flux Density



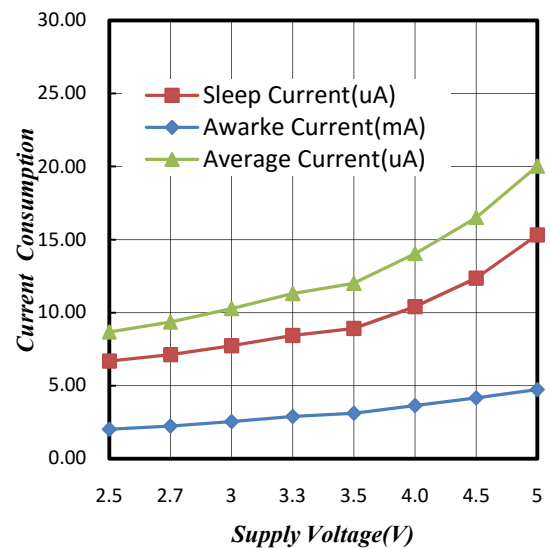
Typical Temperature(T_A) Versus Flux Density



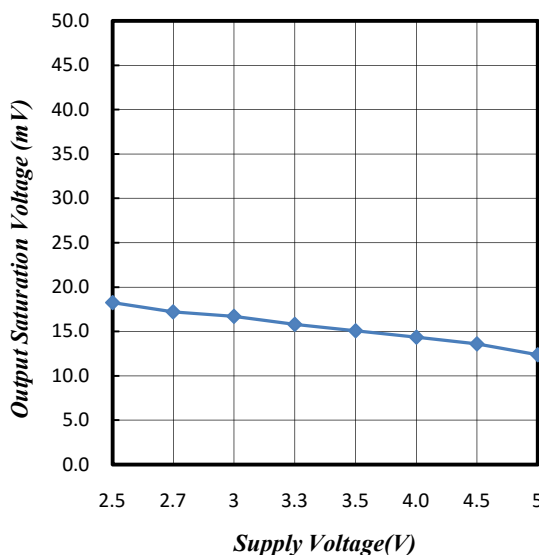
Typical Temperature(T_A) Versus Supply Current(I_{DD})



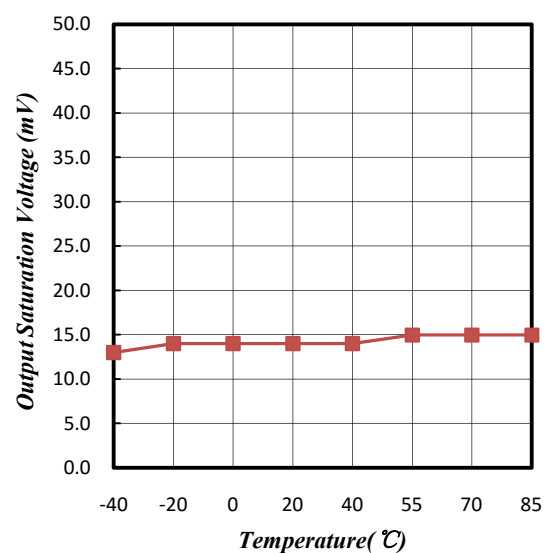
Typical Supply Voltage(V_{DD}) Versus Supply Current(I_{DD})



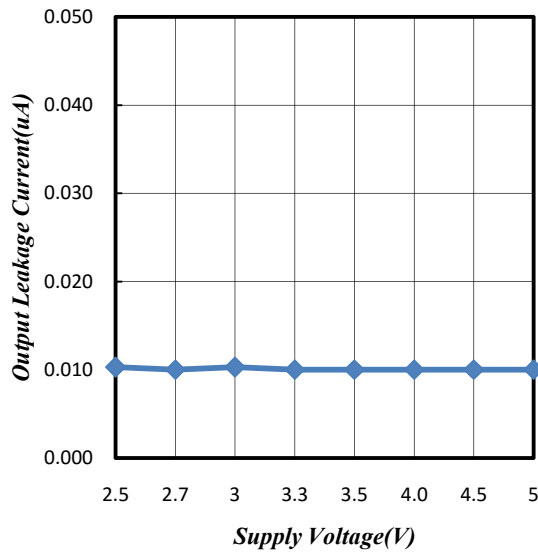
Typical Supply Voltage(V_{DD}) Versus Output Voltage($V_{DS(on)}$)



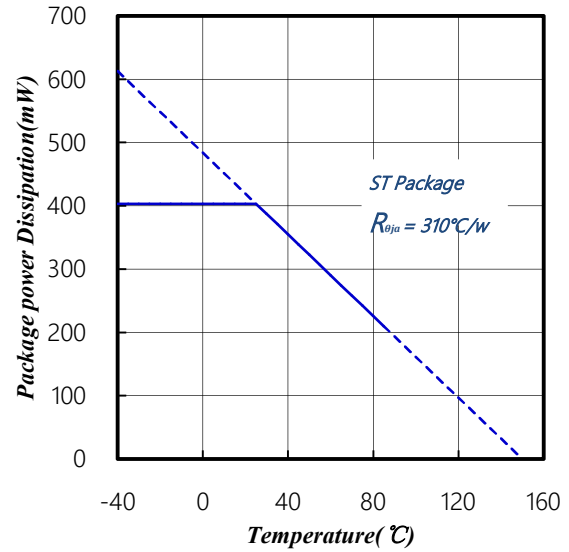
Typical Temperature(T_A) Versus Output Voltage($V_{DS(on)}$)



Typical Supply Voltage (V_{DD}) Versus Leakage Current (I_{OFF})



Power Dissipation versus Temperature (T_A)



Package Power Dissipation

The power dissipation of the Package is a function of the pad size. This can vary from the minimum pad size for soldering to a pad size given for maximum power dissipation. Power dissipation for a surface mount device is determined by $T_{J(max)}$, the maximum rated junction temperature of the die, $R_{\theta JA}$, the thermal resistance from the device junction to ambient, and the operating temperature, T_a . Using the values provided on the data sheet for the package, PD can be calculated as follows:

$$P_D = \frac{T_{J(max)} - T_a}{R_{\theta JA}}$$

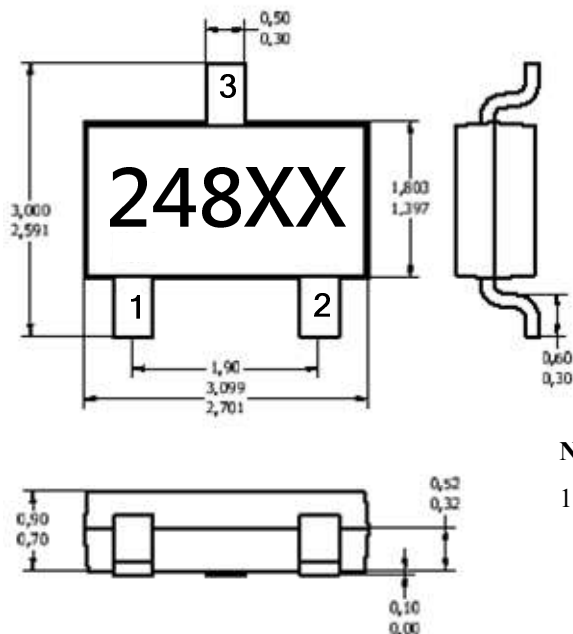
The values for the equation are found in the maximum ratings table on the data sheet. Substituting these values into the equation for an ambient temperature T_a of 25°C, one can calculate the power dissipation of the device which in this case is 400milliwatts.

$$P_D(ST) = \frac{150^\circ\text{C} - 25^\circ\text{C}}{310^\circ\text{C}/\text{W}} = 400\text{mW}$$

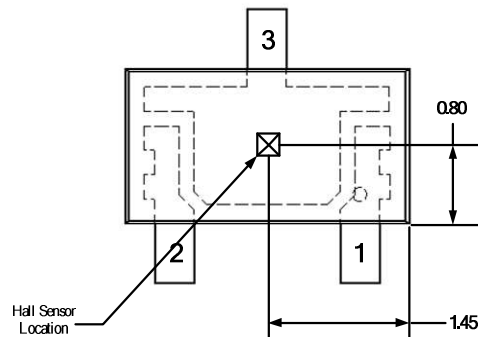
The 310°C/W for the ST package assumes the use of the recommended footprint on a glass epoxy printed circuit board to achieve a power dissipation of 400milliwatts. There are other alternatives to achieving higher power dissipation from the Package. Another alternative would be to use a ceramic substrate or an aluminum core board such as Thermal Clad. Using a board material such as Thermal Clad, an aluminum core board, the power dissipation can be doubled using the same footprint.

Sensor Location, Package Dimension and Marking

ST Package (TSOT-23)
(Top View)



Hall Plate Chip Location
(Bottom view)



NOTES:

- PINOUT (See Top View at left:)
Pin 1 VDD
Pin 2 Output
Pin 3 GND
- Controlling dimension: mm;

MH248REST (TSOT-23) Package Date Code

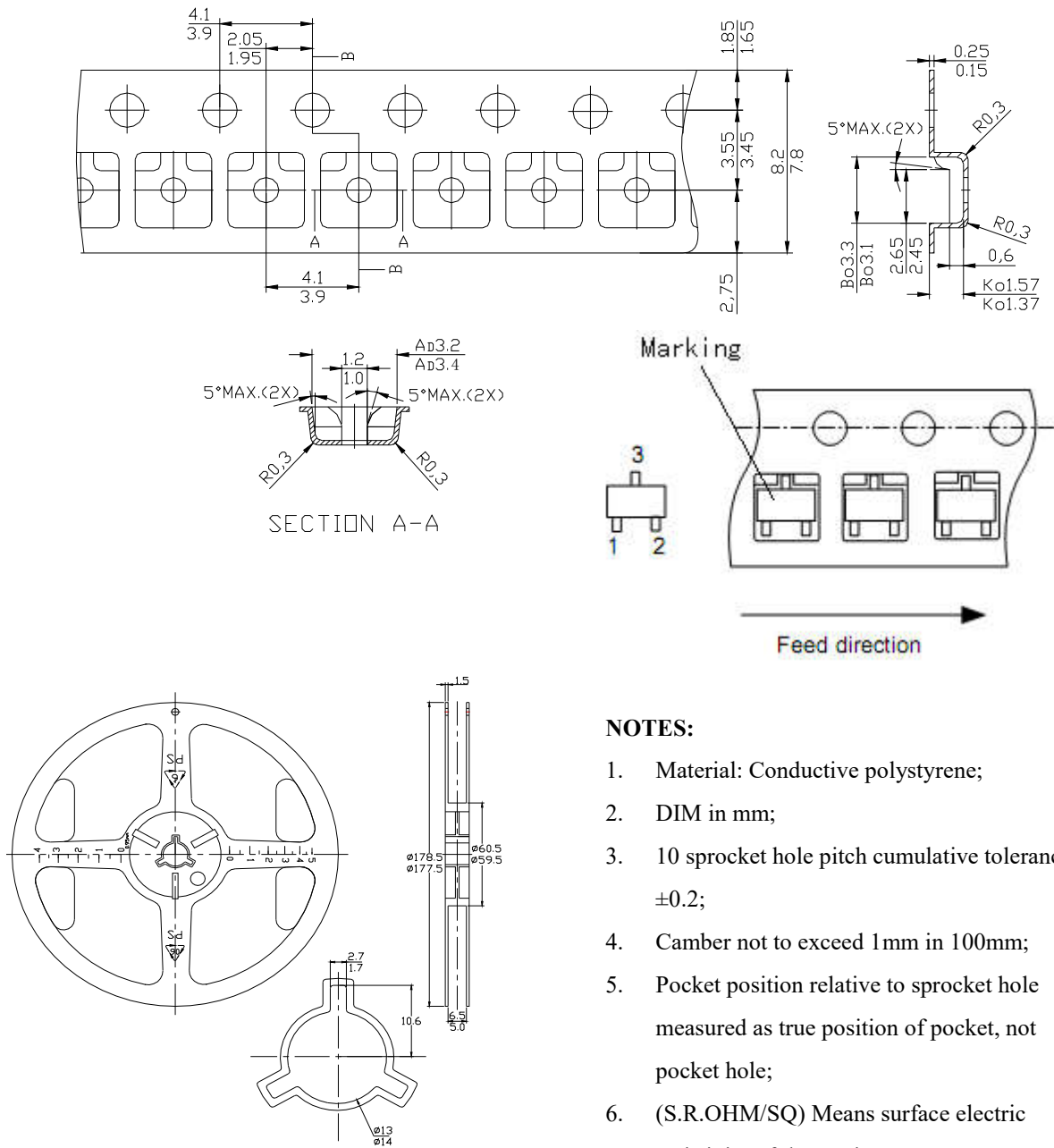
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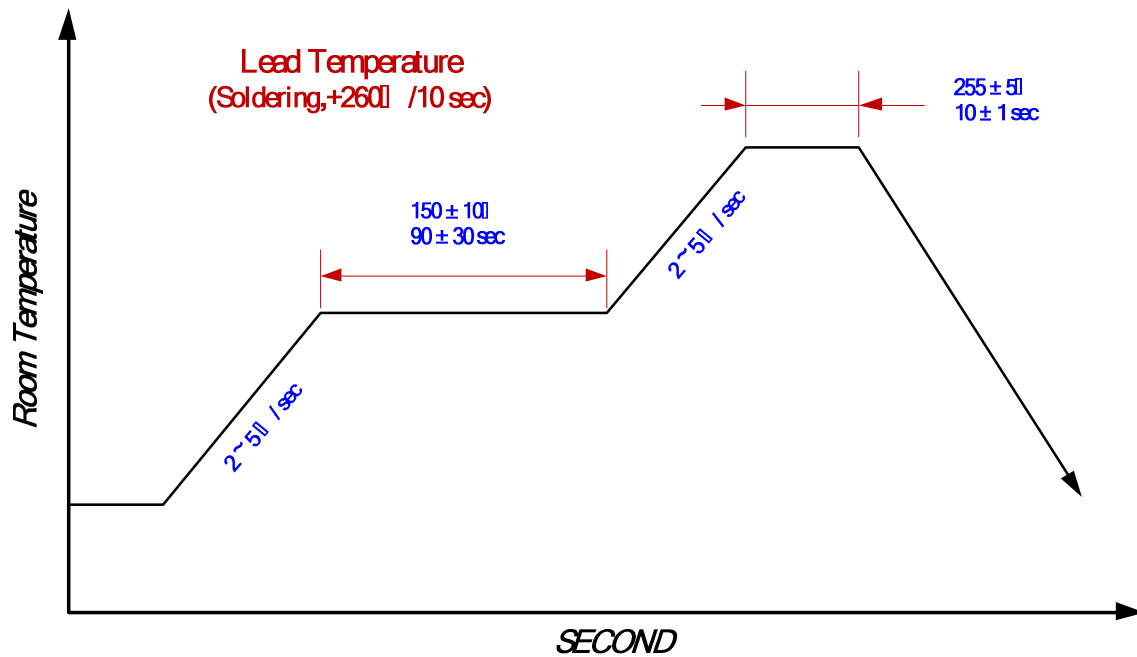
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Week Code

week	1	2	3	4	5	6	7	8	9	10	11	12	13
Code	WA	WB	WC	WD	WE	WF	WG	WH	WI	WJ	WK	WL	WM
week	14	15	16	17	18	19	20	21	22	23	24	25	26
Code	WN	WO	WP	WQ	WR	WS	WT	WU	WV	WW	WX	WY	WZ
week	27	28	29	30	31	32	33	34	35	36	37	38	39
Code	XA	XB	XC	XD	XE	XF	XG	XH	XI	XJ	XK	XL	XM
week	40	41	42	43	44	45	46	47	48	49	50	51	52
Code	WW	WX	WY	WZ	XR	XS	XT	XU	XV	XW	XX	XY	XZ

EX : 2016 Year_8 Week → WH

TSOT-23 package Tape On Reel Dimension





ST Soldering Condition

Packing specification:

Package	Bag	Box	Carton
TSOT-23-3L	3,000pcs/reel	10reel/box	2box/carton

Package	3,000pcs/reel	10reel/box	2box/carton
Weight	0.18kg	1.99kg	4.9kg

TSOT Package Inner box label : Size:3.4cm*6.4cm

Bag and inner box Halogen Free Label



TSOT Carton label : Size:5.6cm* 9.8cm
Bag and inner box Halogen Free Label

 MSL1 260°C	DEVICE : QUANTITY : CARTON OF
 MST Magnesensor Technology	Magnesensor Technology Ltd.

Combine:

When combine lot, one reel could have two D/C and no more than two DC. One carton could have two devices, no more than two;